Firm Life Cycle, Endogenous Production Networks, and Frictional Labor Market

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Disclaimer: The views expressed here are our own and do not necessarily reflect the views of the National Bank of Belgium.

Motivation

- How do firms grow in supply chains?
  - most firms are born small $\rightarrow$ need to find customers, suppliers, and employees to expand
  - face frictions in product market and labor market $\rightarrow$ they potentially interact with each other
- Most of existing frameworks focus on friction in only one market
  - endogenous firm-to-firm trade: often static, frictionless labor market
  - firm dynamics + frictional labor market: often homogeneous final output, no input market
- Yet, unified framework is a key for understanding ...
  - bottleneck of firm growth, policies (labor vs industrial) to promote growth etc.
- **This paper:** characterize firm dynamics in the presence of frictional firm-to-firm trade and worker-firm matching

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What we do...

- **Empirics:** use Belgian data to document how firms expand in product & labor markets
  - universe of firm-to-firm linkages from VAT records (intensive margins + long panel)
  - decompose the margins of firm growth
- **Model:** endogenous production network + search and matching
  - continuous-time firm dynamics model with three margins of adjustment
  - firms vacancies in the labor market and post ads in the product market
- **Application** (not today): use estimated model to evaluate various mixes of policies to reduce vacancy/ads costs

What we find...

- older firms are larger and have more connections
- selection vs life cycle
  - firms that survive are larger from the beginning but also grow over time
  - life cycle of firm-to-firm links: transactions depreciate and get replaced by new links
- decomposition of firm size growth: substantial churning
- firms expand in both labor market and input market

- model implication:
  - both employment and network structure matter for vacancy posting
  - policies may be ineffective if firms are in inaction regions

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Related literature

- Firm life cycle and growth
  - Hsieh and Klenow (2014), Sterk, Sedlácek, and Pugsley (2021), Einav, Klenow, Levin, and Murciano-Goroff (2022), Afrouzi, Drenik, and Kim (2023), Argente, Fitzgerald, Moreira, and Priolo (2024)
- Endogenous production networks
  - Demir, Fieler, Xu, and Yang (2023), Dhyne, Kikkawa, Kong, Mogstad, and Tintelnot (2023), Miyauchi (2024), Arkolakis, Huneeus, and Miyauchi (2025), Aekka and Khanna (2025), Asai and Nirei (2025)
- Firm-to-firm trade + imperfect competition in labor market
  - Dhyne, Kikkawa, Komatsu, Mogstad, and Tintelnot (2022), Huneeus, Kroft, and Lim (2023), Dhyne and Komatsu (2023)
- Firm dynamics + frictional labor market
  - Elsby and Michaels (2013), Coşar, Guner, Tybout (2016), Kaas and Kimasa (2021), Bilal, Engbom, Mongey, and Violante (2022), Elsby and Gottfries (2021)

Data

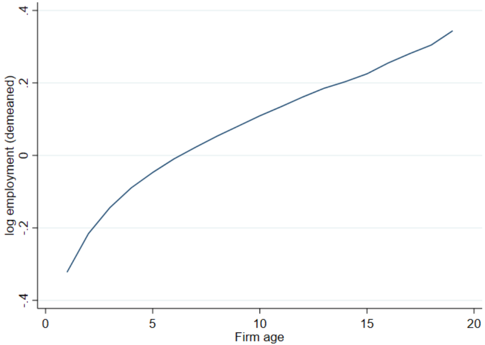
Data sources

- NBB B2B transaction database: panel of VAT-id to VAT-id transactions among the universe of Belgian firms
  - seller id, buyer id, transaction amount
  - low reporting threshold: 250 EUR per year
  - long panel: over years 2002-2022
  - can be merged with NBB annual accounts
    - firm-level information (value added, employment, industry etc.)

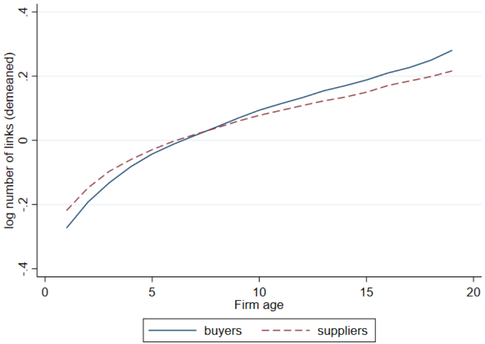
Motivating Empirical Evidence

- **older firms are larger and have more buyers/suppliers**

Age vs employment



Age vs number of buyers/suppliers



- pooled cross-section in logs over 2002-2022
- demeaned using four-digit industry-year fixed effects

Life cycle of firm growth

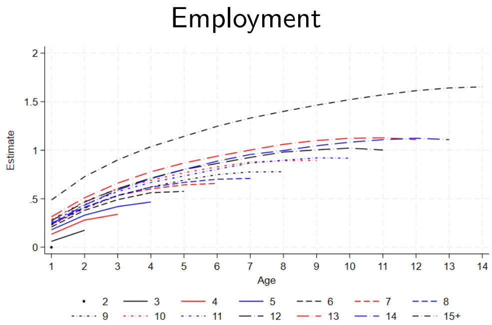
▸ + industry-year FE ▸ manufacturing ▸ utility&construction ▸ retail&wholesale ▸ service

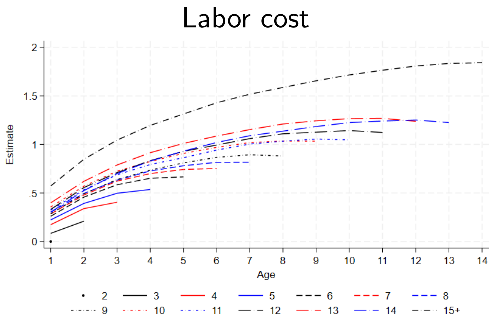
- **firm growth vs selection of surviving firms?**
- show the life cycle patterns of firm-level outcomes (à la Argenete et al., 2024)

$$y_{j,t} = \alpha + \sum_{d=2}^{\bar{a}} \sum_{a=1}^{d-1} \beta_a^d 1_{\{t=c(j)+a, D(j)=d\}} + \gamma_{c(j)} + \tau_t + \epsilon_{j,t}$$

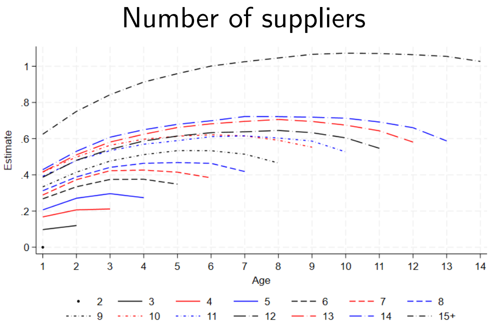
  - $y_{j,t}$: log employment, labor cost, number of suppliers, network inputs, number of buyers, and network sales
  - $c(j)$: firm j's cohort based on the year before its first year of full activity
  - $D(j)$: firm j's duration, i.e. maximum age
  - $\gamma_{c(j)}$: cohort effect
  - τ_t: calendar year fixed effect

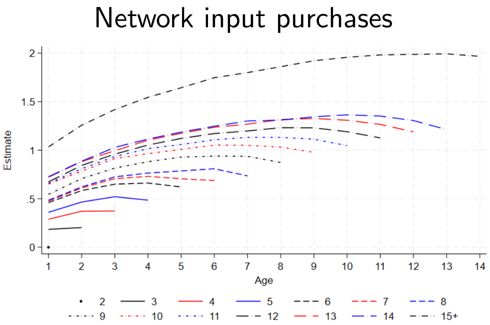
Life cycle of firm growth: labor market





Life cycle of firm growth: input market





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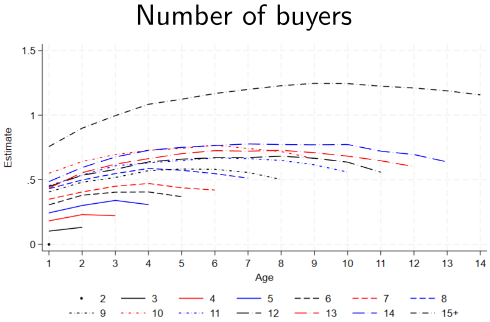
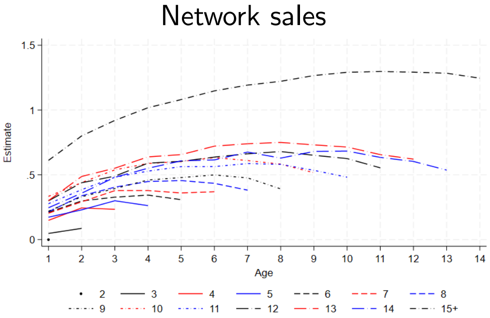


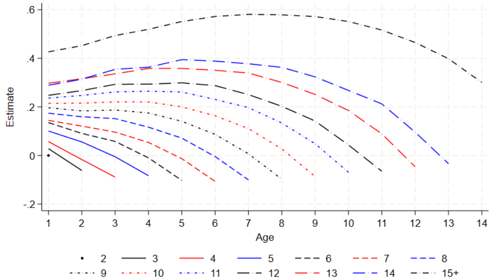


Life cycle of firm-to-firm transactions

- link age vs firm-to-firm transactions

$$y_{i,j,t} = \alpha + \sum_{d=2}^{\bar{a}} \sum_{a=1}^{d-1} \beta_a^d 1_{\{t=c(i,j)+a, D(i,j)=d\}} + \gamma_{c(i,j)} + \tau_t + \epsilon_{i,j,t}$$

- average links depreciate over time as relationship ages



Decomposing firm size growth

summary statistics

- **what are the sources of firm size growth?**
- consider the following exact decomposition of network sales growth:

$$\frac{x^B_{j,t+1} - x^B_{j,t}}{x^B_{j,t}} = \frac{|B_{j,t+1}|\bar{x}^B_{j,t+1} - |B_{j,t}|\bar{x}^B_{j,t}}{|B_{j,t+1}|\bar{x}^B_{j,t}}$$

  - B: set of buyers; $\bar{x}$: average sales per buyer
  - Growth from continuing links and link churning
  - Extensive (number of links) and intensive (trade volume per link) margins
  - Aggregate firm-level growth using firm size as weights
- Similar decomposition for network input growth

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Decomposing size growth: main results

	Total	Continuing	New	Exiting
Panel A: Network Sales Growth				
Total	0.019	0.012	0.083	-0.075
Extensive margin		0.680	0.323	-0.287
Intensive margin		0.044	0.273	0.230
Panel B: Network Input Growth				
Total	0.021	0.013	0.085	-0.073
Extensive margin		0.664	0.312	-0.301
Intensive margin		0.014	0.245	0.220

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- $\sim 2\%$ growth: substantial contribution from link churning
- $\sim \frac{1}{3}$ links are replaced by more new links with higher trade volumes per link

Decomposing network sales growth: results by size quintile network input

Quintile:	(1)	(2)	(3)	(4)	(5)
Total	1.102	0.286	0.095	0.060	0.013
Continuing	0.210	0.090	0.031	0.016	0.011
Extensive	0.367	0.456	0.540	0.595	0.690
Intensive	0.414	0.184	0.054	0.026	0.044
New	1.059	0.441	0.248	0.181	0.070
Extensive	0.563	0.457	0.405	0.384	0.316
Intensive	1.328	0.756	0.513	0.429	0.255
Exiting	-0.374	-0.272	-0.186	-0.140	-0.067
Extensive	-0.441	-0.400	-0.356	-0.340	-0.282
Intensive	0.556	0.487	0.388	0.327	0.219

- Growth of smaller (larger) firms is driven more by link churning (continuing links)
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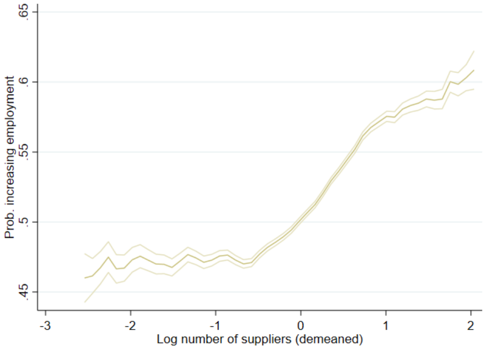
- Growth of smaller (larger) firms is driven more by link churning (continuing links)
- Link churning is stronger for smaller firms at both extensive and intensive margins

decomposing size growth: additional results

- link churning: industry churning vs firm churning within industry
  - Industry churning is more important for smaller firms than firm churning within industry
    Network input Network sales
- main results are robust using various subsamples
  - young vs old firms Results
  - sector Results

- **expansion in product vs labor markets: substitutes or complements?**
- consider number of suppliers demeaned by industry $\times$ age FE

prob. of adding employee



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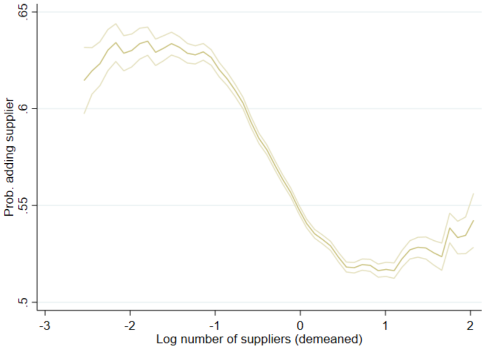

Model

Model: environment

- continuous time, discount rate at ρ
- measure L of workers, set of infinitesimally small firms $j \in \Omega_t$
- each firm j produces uniquely differentiated good
    - combines labor inputs and intermediate inputs (from $\Omega^S_{j,t}$)
- at each moment ...
    - static problem
    - dynamic problem

$\rightarrow$ consider aggregate steady state, no idiosyncratic productivity shocks

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- each firm j produces uniquely differentiated good
- at each moment ...
  - static problem
    - firm j takes as given the number of workers and network structure
    - firm and incumbent workers bargain over their wages
    - firm sets a price and sell its good to household and customer firms (from $\Omega^B_{j,t}$)
  - dynamic problem

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    - firm posts vacancy in the labor market to hire workers
    - firm posts ads to search for buyers and suppliers

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Model: consumption decision

- consumption: CES preference over differentiated goods:

$$U = \left(\int_{\Omega} q_{jH}^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}}$$

  - Ω: set of firms in the economy
  - demand for goods produced by firm j, q_{jH}:

$$q_{jH} = \frac{p_{jH}^{-\sigma}}{P^{1-\sigma}} E \quad \text{where} \quad P = \left(\int_{\Omega} p_{kH}^{1-\sigma} dk \right)^{\frac{1}{1-\sigma}}$$

- Cobb-Douglas in labor and intermediate inputs

$$q_j = \phi_j n_j^{\alpha} m_j^{1-\alpha}$$

  - labor inputs: n_j
  - intermediate inputs: $m_j = \left(\int_{\Omega_j^S} q_{ij}^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}}$
    - assumption: elasticity of substitution σ is the same as in preference

- Demand for q_{ij} when purchasing m_j:

$$q_{ij}(m) = p_{ij}^{-\sigma} z_j^{\frac{\sigma}{1-\sigma}} m \quad \text{where} \quad z_j = \int_{\Omega_j^S} p_{kj}^{1-\sigma} dk$$

- Suppliers set prices for intermediate inputs
- **Result:** Firm j sets $p_{jk} = p_j \quad \forall k \in \Omega_j^B \cup \{H\}$
    - facing demand

$$q_j \equiv \int_{\Omega_j^B \cup \{H\}} q_{jk} dk = \chi_j p_j^{-\sigma},$$

$$\chi_j = \int_{\Omega_j^B} z_k^{\frac{\sigma}{1-\sigma}} m_k dk + P^{\sigma-1} E$$

Model: revenue maximization problem

- **Static sub-problem:** revenue net of intermediate input costs ...
  - taking as given employment and network structure

$$R(\phi, n, \Omega^B, \Omega^S) = \max_{p,m} \left(pq - z^{\frac{1}{1-\sigma}} m \right) \quad \text{s.t.} \quad q = \phi n^\alpha m^{1-\alpha} \ \text{ and } \ q = \chi p^{-\sigma}$$

  - closed-firm solution

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  - return to labor inputs depends on ϕ, χ, and z

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Model: dynamic problem, labor market

- firms post vacancies v^ℓ with vacancy cost $c^\ell(v^\ell)$
- matching: standard CRS matching function

$$M^L\Big(\underbrace{L - \int_\Omega n d\omega}_{\text{job searcher}\equiv U},\ \underbrace{\int_\Omega v^\ell d\omega}_{\text{total vacancy}\equiv V^\ell}\Big)$$

  - $\omega \equiv (\phi, n, \chi, z)$
  - job finding rate $\lambda = M^L/U$, vacancy filling rates $\mu^\ell = M^L/V^\ell$
- wage determined by Nash bargaining over marginal surplus (Stole-Zwiebel, 1996, and Elsby-Michaels, 2013)
- incumbent workers are exogenously separated at rate δ^ℓ

Model: dynamic problem, product market

- firms post ads v^u to search for buyers and v^d to search for suppliers
  - ads costs $c^u(v^u)$ and $c^d(v^d)$
- matching function

$$M^P(\underbrace{\int_\Omega v^u d\omega}_{\equiv V^u}, \underbrace{\int_\Omega v^d d\omega}_{\equiv V^d})$$

  - ad filling rates $\mu^u = M^P/V^u$ and $\mu^d = M^P/V^d$
- firms' current χ and z depreciate at rates $\delta^u(\chi)$ and $\delta^d(z)$

Model: dynamic problem, product market (cont'd)

- $\chi'(\chi, v^u)$: expected demand shifter when a firm with current χ posts v^u ads

  $\implies$ by the Reynolds Transport Theorem:

$$\frac{\partial \chi'}{\partial v^u} = \mu^u \mathbb{E}_{v^d}\left[z^{\frac{\sigma}{1-\sigma}} m\right]$$

$$\text{where} \quad m(\omega) = \left[\frac{(1-\alpha)(\sigma-1)}{\sigma} z_j^{\frac{1}{\sigma-1}} \phi^{\frac{\sigma-1}{\sigma}} \chi^{\frac{1}{\sigma}} n^{\frac{\alpha(\sigma-1)}{\sigma}}\right]^{\frac{\sigma}{1+\alpha(\sigma-1)}}$$

- similarly for $z'(z, v^d)$:

$$\frac{\partial z'}{\partial v^d} = \mu^d \mathbb{E}_{v^u}\left[p^{1-\sigma}\right]$$

$$\text{where} \quad p(\omega) = \left[\left(\frac{\sigma}{(1-\alpha)(\sigma-1)}\right)^{1-\alpha} z^{\frac{1-\alpha}{1-\sigma}} \phi^{-1} \chi^{\alpha} n^{-\alpha}\right]^{\frac{1}{1+\alpha(\sigma-1)}}$$

- Focus on firm's problem at the steady state
- **Firm's value function:** firm chooses how many vacancies and ads to post (v^ℓ, v^u, v^d):

$$\begin{aligned}\rho\Pi(\omega) = \max_{v^\ell\geq 0, v^u\geq 0, v^d\geq 0} \Bigg[& R(\omega) - w(\omega)n \\ & - c^\ell(v^\ell) + \left(\mu^\ell v^\ell - \delta^\ell n\right)\frac{\partial \Pi}{\partial n} \\ & - c^u(v^u) + \left(\chi'(\chi, v^u) - \chi - \delta^u(\chi)\right)\frac{\partial \Pi}{\partial \chi} \\ & - c^d(v^d) + \left(z'(z, v^d) - z - \delta^d(z)\right)\frac{\partial \Pi}{\partial z}\Bigg]\end{aligned}$$

Key intuition

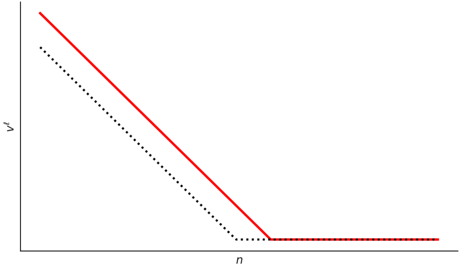
- consider linear vacancy cost
- from KKT condition for optimal vacancy:

$$v^\ell(\phi, n, \chi, z) = 0 \quad \text{or}$$

$$\frac{c^\ell}{\mu^\ell} = \frac{\partial \Pi \left(\phi, n + \mu^\ell v^\ell(\phi, n, \chi, z), \chi, z\right)}{\partial n}$$

- lower ϕ, χ, z $\rightarrow$ greater inaction region
- policy to reduce vacancy cost may not be effective if firms are in the inaction region

Optimal $v^\ell(\phi, n, \chi, z)$: fixed ϕ, χ, z

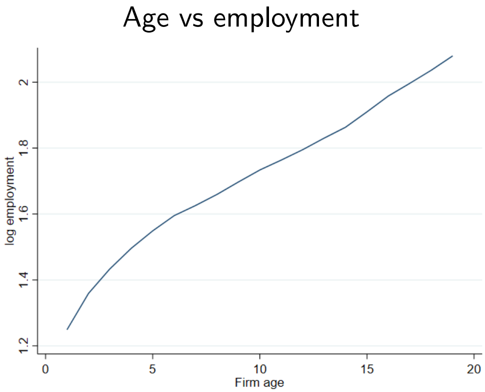


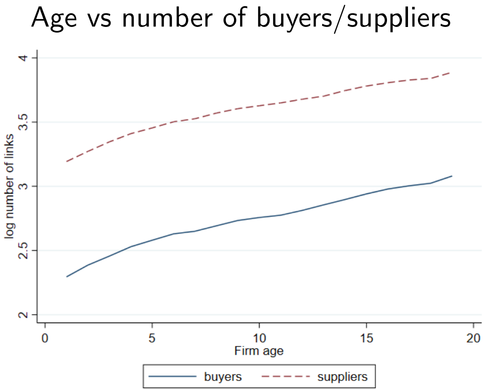
Conclusion and next steps

- **quantitative questions:** given the equilibrium distribution of firms...
  - are policies to reduce frictions in labor and/or product markets effective?
  - which firms are likely to be affected?
  - $\rightarrow$ consider various profiles of labor + industrial policy mixes

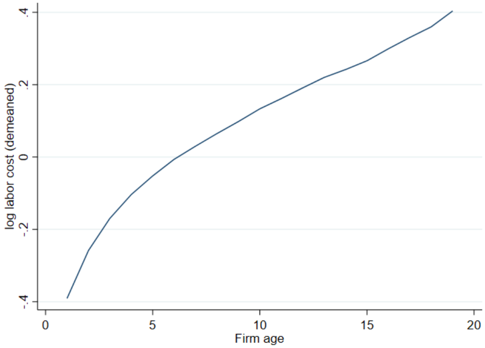
- firms grow in both product and labor markets
- life cycle of firm-to-firm transactions and churning of links
- need a unified framework to consider frictions in both markets simultaneously

Appendix

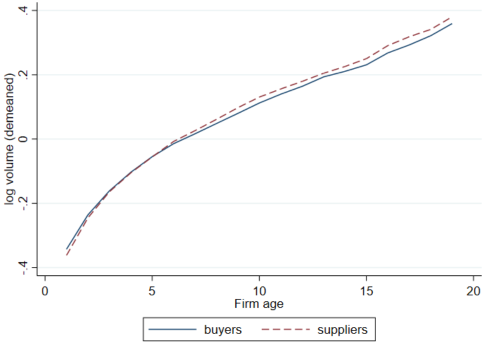

Age vs employment
log employment
1.2
1.4
1.6
1.8
2
0
5
10
15
20
Firm age


Age vs number of buyers/suppliers
log number of links
2
2.5
3
3.5
4
0
5
10
15
20
Firm age
buyers
suppliers

Age vs labor cost



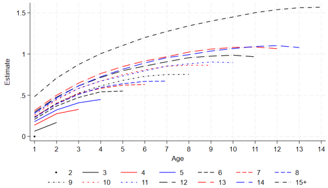
Age vs network sales/inputs



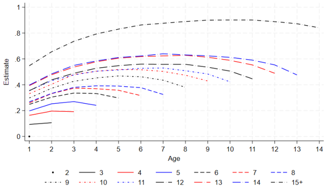
Life cycle of firm growth: + industry-year FE

back

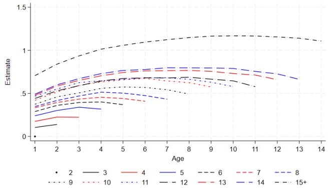
Employment



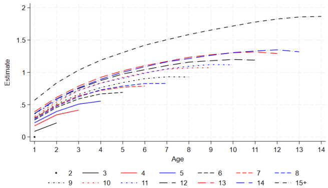
Number of suppliers



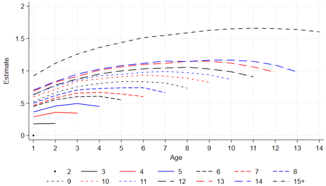
Number of buyers



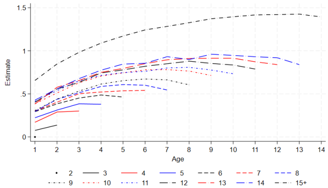
Labor cost



Network inputs



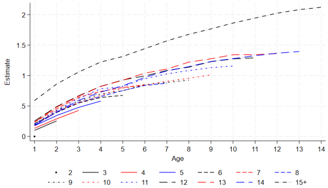
Network sales



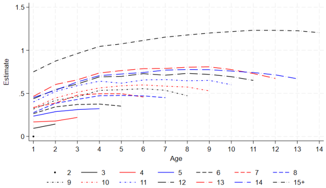
Life cycle of firm growth: manufacturing

back

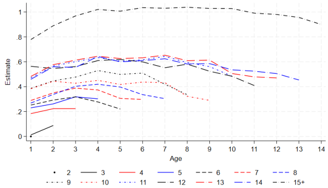
Employment



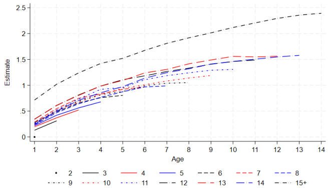
Number of suppliers



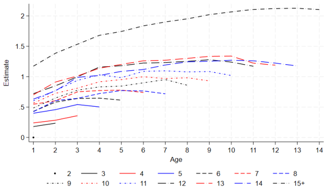
Number of buyers



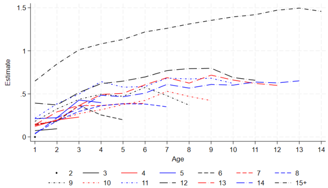
Labor cost



Network inputs



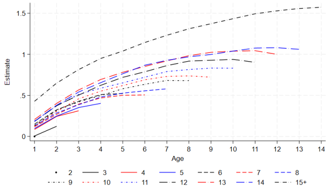
Network sales



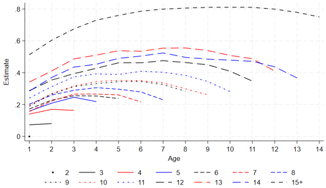
Life cycle of firm growth: utility and construction

back

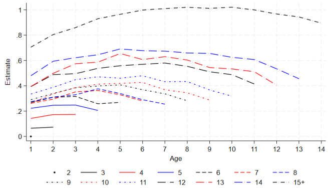
Employment



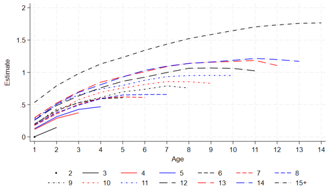
Number of suppliers



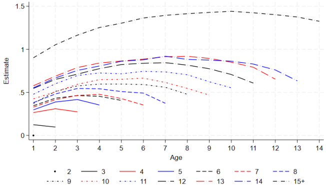
Number of buyers



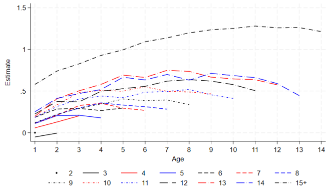
Labor cost



Network inputs



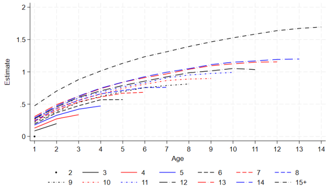
Network sales



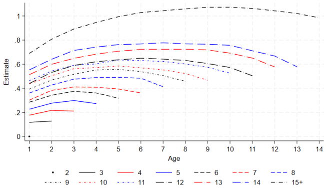
Life cycle of firm growth: retail and wholesales

back

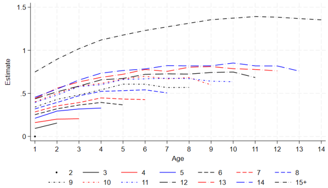
Employment



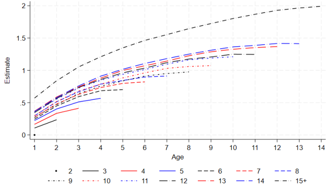
Number of suppliers



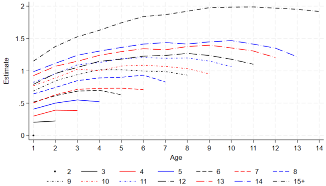
Number of buyers



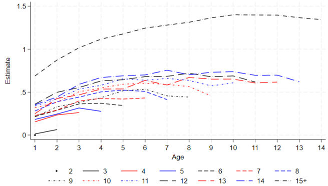
Labor cost



Network inputs

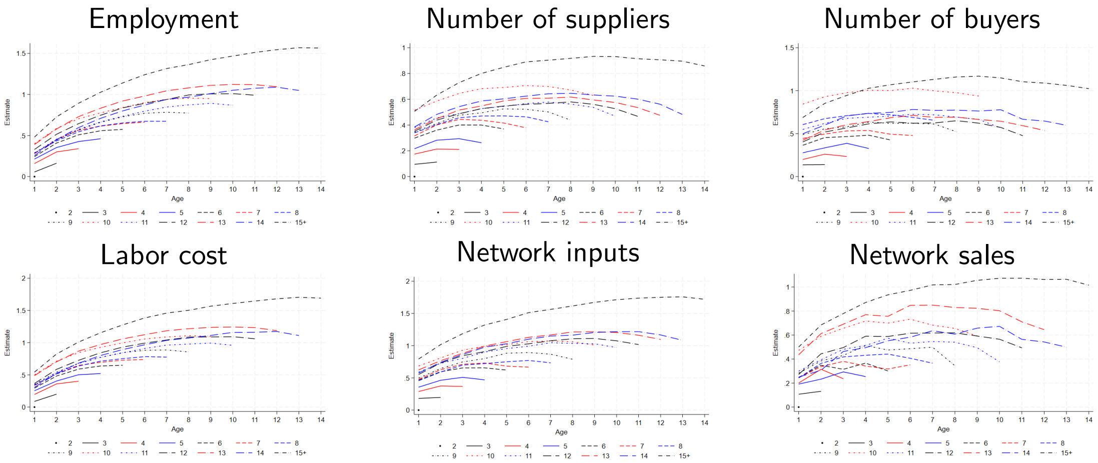


Network sales



Life cycle of firm growth: service

back



Summary Statistics

- Sample period: 2012-2016

	All firms	By size quintile				
		(1)	(2)	(3)	(4)	(5)
Total sales, million	7.08	0.16	0.39	0.76	1.72	32.36
Network sales, million	2.66	0.07	0.15	0.29	0.73	11.12
Outdegree	50.96	7.13	13.91	24.86	47.70	161.21
Outdegree, NACE4 industry	17.52	4.94	8.77	13.45	20.87	39.55
Network sales per buyer, million	0.13	0.03	0.03	0.05	0.08	0.42
Network input, million	2.38	0.11	0.18	0.36	0.81	10.44
Indegree	50.96	16.95	26.56	37.13	52.71	121.47
Indegree, NACE4 industry	28.16	12.72	18.72	24.09	31.12	54.16
Network input per supplier, million	0.02	0.01	0.01	0.01	0.02	0.08
Labor costs, thousand	977.07	93.96	115.67	197.24	405.73	4072.74
Employment	17.28	2.45	3.07	4.83	8.92	67.12
Average wage, thousand	44.33	35.47	38.56	42.29	47.08	58.24

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Decomposing Network Input Growth: Results by Size Quintile

Quintile:	(1)	(2)	(3)	(4)	(5)
Total	0.377	0.138	0.072	0.036	0.016
Continuing	0.079	0.035	0.015	0.008	0.012
Extensive	0.481	0.536	0.579	0.618	0.672
Intensive	0.132	0.051	0.018	0.006	0.014
New	0.454	0.270	0.208	0.149	0.070
Extensive	0.369	0.360	0.358	0.342	0.307
Intensive	0.875	0.607	0.507	0.388	0.219
Exiting	-0.207	-0.174	-0.153	-0.121	-0.065
Extensive	-0.320	-0.334	-0.340	-0.326	-0.299
Intensive	0.476	0.427	0.393	0.329	0.203

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Decomposing Network Input Growth: Industry Churning

Quintile:	(1)	(2)	(3)	(4)	(5)
New firm, new industry	0.307	0.155	0.102	0.060	0.013
New firm, cont. industry	0.125	0.110	0.103	0.088	0.056
Cont. firm, cont. industry	0.079	0.035	0.015	0.008	0.012
Exiting firm, cont. industry	-0.064	-0.074	-0.076	-0.072	-0.052
Exiting firm, exiting industry	-0.142	-0.099	-0.076	-0.048	-0.012

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Decomposing Network Sales Growth: Industry Churning

Quintile:	(1)	(2)	(3)	(4)	(5)
New firm, new industry	0.827	0.285	0.132	0.075	0.015
New firm, cont. industry	0.152	0.141	0.113	0.104	0.055
Cont. firm, cont. industry	0.210	0.090	0.031	0.016	0.011
Exiting firm, cont. industry	-0.063	-0.088	-0.084	-0.080	-0.051
Exiting firm, exiting industry	-0.309	-0.183	-0.101	-0.060	-0.015

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Decomposing Network Sales Growth: By Size Quintile and Age Group

Size quintile:	(1)		(2)		(3)		(4)		(5)	
Age:	Young	Old	Young	Old	Young	Old	Young	Old	Young	Old
Total	1.207	1.058	0.235	0.185	0.100	0.046	0.070	0.024	0.027	-0.002
Continuing	0.183	0.141	0.074	0.054	0.034	0.007	0.026	-0.004	0.026	-0.002
Extensive	0.350	0.377	0.436	0.467	0.528	0.546	0.585	0.599	0.695	0.690
Intensive	0.361	0.284	0.139	0.107	0.052	0.005	0.031	-0.016	0.030	-0.014
New	1.328	1.226	0.414	0.382	0.242	0.218	0.178	0.164	0.071	0.066
Extensive	0.601	0.569	0.487	0.436	0.436	0.384	0.414	0.365	0.422	0.285
Intensive	1.440	1.436	0.652	0.695	0.450	0.477	0.373	0.400	0.207	0.229
Exiting	-0.372	-0.375	-0.275	-0.271	-0.184	-0.187	-0.138	-0.140	-0.072	-0.065
Extensive	-0.435	-0.445	-0.397	-0.401	-0.349	-0.358	-0.331	-0.342	-0.267	-0.282
Intensive	0.546	0.563	0.477	0.494	0.370	0.396	0.312	0.330	0.205	0.218

- Young firms: firm age is no greater than 10 years old

Back

Decomposing Network Sales Growth: Sector

Sector:	Agricultural Mining	Manufacturing	Retail Wholesale	Service	Utility Construction
Total	0.016	-0.012	0.005	0.034	0.109
Continuing	0.016	-0.008	0.002	0.024	0.069
Extensive	0.662	0.698	0.698	0.657	0.622
Intensive	0.027	-0.024	0.005	0.084	0.256
New	0.091	0.061	0.076	0.093	0.135
Extensive	0.317	0.278	0.272	0.337	0.576
Intensive	0.239	0.260	0.279	0.274	0.280
Exiting	-0.092	-0.063	-0.072	-0.082	-0.094
Extensive	-0.272	-0.281	-0.270	-0.304	-0.334
Intensive	0.226	0.197	0.250	0.235	0.210

Back

- Karush-Kuhn-Tucker conditions for optimal vacancies ($v^\ell \geq 0$) and ads ($v^u, v^d \geq 0$)

$$v^{\ell *}\left(\frac{dc^\ell}{dv^\ell} - \mu^\ell \frac{\partial \Pi}{\partial n}\right) = 0$$

$$v^{u*}\left(\frac{dc^u}{dv^u} - \frac{\partial \chi'}{\partial v^u}\frac{\partial \Pi}{\partial \chi}\right) = 0 \implies v^{u*}\left(\frac{dc^u}{dv^u} - \mu^u \mathbb{E}_{v^d}\left[z^{\frac{\sigma}{1-\sigma}} m\right]\frac{\partial \Pi}{\partial \chi}\right) = 0$$

$$v^{d*}\left(\frac{dc^d}{dv^d} - \frac{\partial z'}{\partial v^d}\frac{\partial \Pi}{\partial z}\right) = 0 \implies v^{d*}\left(\frac{dc^d}{dv^d} - \mu^d \mathbb{E}_{v^u}\left[p^{1-\sigma}\right]\frac{\partial \Pi}{\partial z}\right) = 0$$

back

- Policy functions: $v^\ell(\omega,t), v^u(\omega,t), v^d(\omega,t)$, for $\omega = (\phi, n, \chi, z)$
- Law of motion:

$$\begin{aligned} b_n(\omega,t) &= \mu^\ell v^\ell(\omega,t) - \delta^\ell n \\ b_\chi(\omega,t) &= \chi'(\chi, v^u(\omega,t)) - \chi - \delta^u(\chi) \\ b_z(\omega,t) &= z'(z, v^d(\omega,t)) - z - \delta^d(z) \end{aligned}$$

- Kolmogorov forward equation for density $g(\omega,t)$:

$$\partial_t g(\omega,t) = -\partial_n\big(b_n(\omega,t)g(\omega,t)\big) - \partial_\chi\big(b_\chi(\omega,t)g(\omega,t)\big) - \partial_z\big(b_z(\omega,t)m(\omega,t)\big)$$